Chapter 14

General Agents

Working with Claude Code and Cowork

Panaversity — General Agents Foundations

Complete Study Notes

Prepared for: Imran Hassan | 2026

Chapter Overview

Yeh chapter General Agents practically use karna sikhata hai — especially Claude Code aur Claude Cowork. Understanding se experience tak ka safar.

> **Opening Quote:** "You know why AI-driven development matters. But understanding transformation and experiencing it are two different things."

General Agent kya hai?

An AI system that **observes, orients, decides, and acts** via the OODA loop — sirf text generate nahi karta, actual actions execute karta hai.

General Agent (Claude Code)

- Novel problems solve karta hai
- Instant setup
- Diverse tasks
- Senior consultant ki tarah
- **"Builders hain"**

Custom Agent (SDK se)

- Specific repetitive workflow
- Weeks of design
- Customer-facing solutions
- Factory machine ki tarah
- **"Products hain"**

Chapter Structure — 5 Sections

Section	Topic	Lessons
A	Claude Code Essentials	1–7
B	Skills, Subagents & MCP	8–14
C	Extensibility & Teams	15–25
D	Claude Cowork	25–35
E	Strategy & Assessment	37–41

Section A: Claude Code Essentials (Lessons 1–7)

Lesson 1: Claude Code Origin Story

Date	Event
Sep 2024	Boris Cherny ne Claude ko filesystem access diya (experiment)
Nov 2024	Internal dogfooding — Day 1: 20%, Day 5: 50% adoption
May 2025	Public launch — 80%+ engineers use daily; PR throughput +67%
Mid 2025	Revenue $500M+ annually (word-of-mouth)
Jan 2026	Claude Cowork launched for knowledge workers

> **Product Overhang:** Latent capability already in a model — sirf sahi product interface ki zarurat thi unlock karne ke liye. Claude Code + Cowork dono product overhangs the.

Passive AI

User problem batata hai → generic suggestions milti hain → copy-paste karta hai → test karta hai. AI code kabhi nahi dekhta.

Agentic AI (Claude Code)

User goal batata hai → Claude files padhta hai → changes propose karta hai → tests run karta hai → errors dekh ke iterate karta hai.

The 8 Modes of Claude (2026)

Mode	Function
Chat	Conversational Q&A
Cowork	Local machine files par kaam karta hai
Projects	Persistent workspace (prompts, files, context)
Artifacts	Interactive apps/dashboards within chat
Excel	Spreadsheet integration
Connectors	Gmail, Drive, Slack, Notion, Figma, 50+ services

Plugins	Pre-built agent packs (sales, marketing, legal)
Skills	Reusable task templates with specialist knowledge

Lesson 2: Installation & Authentication

> **Important:** Free Claude.ai plan mein Claude Code nahi milta. Minimum **Pro subscription ($20/month)** chahiye!

3 Authentication Methods:

1. **Claude App Auth** — Pro/Max/Team subscribers ke liye (browser redirect)
2. **Console API Auth** — Pay-per-use ke liye (API key)
3. **Enterprise Auth** — Amazon Bedrock, Google Vertex AI

Diagnostic Command:

```
claude doctor    ← installation, auth, compatibility check karta hai
```

Lesson 3: Free Backends

Backend	Cost	Setup
OpenRouter	Daily limits	Manual
Gemini 2.5 Flash	Daily limits	Easiest
DeepSeek	~$0.028-0.42/million tokens	Slightly complex
Groq	Free tier	Manual

> **Key Point:** Saare features identically kaam karte hain free backends se — Skills, MCP, Subagents, Hooks sab!

Lesson 4: Hello Claude — Approval Pattern

Action	Approval Needed?
Files padhna (read-only)	Nahi — automatically hota hai
Files likhna/create karna	Haan — Enter press karo

Terminal commands run karna	Haan — Enter press karo
Reject karna	Escape press karo

Lesson 5: CLAUDE.md Context Files

CLAUDE.md = Persistent markdown file jo Claude Code har session ke start mein automatically load karta hai. LLM stateless hota hai — CLAUDE.md = external memory!

6 Sections of CLAUDE.md:

1. Project Overview
2. Technology Stack
3. Directory Structure
4. Coding Conventions
5. Key Commands (run, test, deploy)
6. Important Notes (gotchas, security)

Create Kaise Karein:

```
/init    ← Claude khud codebase analyze karke CLAUDE.md banata hai
```

Best Practice: Under 200 lines rakho. Troubleshooting: File ka naam exactly `CLAUDE.md` (case-sensitive), project root mein honi chahiye.

File	Writer	Purpose
CLAUDE.md	Developer	Instructions, standards, project context
AGENTS.md	Developer	Universal — saare AI tools ke liye (60,000+ projects)
Auto Memory	Claude	Discovered learnings (~/.claude/projects/)

Best Practice: AGENTS.md — universal content. CLAUDE.md — Claude-specific features (skills, MCP, hooks). CLAUDE.md should reference AGENTS.md.

Section B: Skills, Subagents & MCP (Lessons 8–14)

Lesson 8: The Concept Behind Skills

> **Core Thesis:** "Stop building agents. Build skills instead."

Intelligence + Code + Execution Framework:

Models (Intelligence) + Code (Execution) + Skills (Expertise) = Complete (Digital FTE)

> **"Intelligence + execution still isn't expertise." Skills domain expertise provide karte hain!**

4 Skill Sources:

1. **Bundled Skills** — Built-in: /batch, /debug, /loop, /simplify
2. **Foundational Skills** — Universal utilities
3. **Partner Skills** — Tool-specific (Browserbase, Notion)
4. **Enterprise Skills** — Organization-specific

MCP	Skills
Provides the "what" — data pipes	Encode the "how" — procedures
Agent ke haath (connectivity)	Agent ki training (expertise)

Progressive Disclosure — 3 Levels:

1. Level 1: Brief metadata (~always loaded) — relevance check
2. Level 2: Complete SKILL.md (on-demand)
3. Level 3: Supporting files (if needed)

Lesson 9: Building Your Own Skills

Skills `.claude/skills/` mein store hote hain. Har skill ek `SKILL.md` file chahiye.

SKILL.md Structure:

```
---
name: skill-name
description: When and why to use this skill (ACTIVATION TRIGGER)
---

## When to Use
## Procedure
## Output Format
```

> **Fast Way:** `skill-creator` meta-skill use karo — Claude khud interview karke skill banata hai!

Description Field Formula:

"[Action verb] + [input type] + [into/for] + [output type]. Use when [trigger conditions]."

2 Activation Methods:

- **Automatic:** Claude khud recognize karta hai
- **Explicit:** Skill ka naam le ke invoke karo

Lesson 11: Subagents & Orchestration

Subagent = specialized AI Agent with its own instructions and **isolated context window**.

Built-in Subagents:

Subagent	Responsibility	Model
Explore	Files locate, code search	Haiku
Plan	Complex multi-step tasks	Sonnet
general-purpose	Various tools tasks	—
Bash	Command execution	—
claude-code-guide	Claude Code questions	Haiku

> **Parallel Execution:** Multiple subagents simultaneously execute karte hain!
> "Use Explore to show files AND use Plan to outline README."

Skills vs Subagents:

Skills Use Karo	Subagents Use Karo

Repeated patterns	Complex workflows
Automatic triggers	Guaranteed execution
Lightweight operations	Comprehensive analysis
Shared context OK	Isolated context needed

Lesson 12: MCP Integration

MCP = Claude Code ko local filesystem se bahar extend karta hai — websites, APIs, databases.

> **"Claude Code can only see files on your computer." MCP = "safe, approved access to the outside world."**

Token Management:

- 5 MCP servers = 25,000–40,000 tokens before any query
- Tool Search activates when definitions >10% context
- ~85% automatic overhead reduction

3 Pillars Formula:

```
CLAUDE.md (Context) + Skills (Procedures) + MCP (Access) = Digital FTE
```

Lesson 13: MCP to Skills (Token Optimization)

Approach	Tokens	Savings
Direct MCP	9,500–12,500	Baseline
Compiled Skill	~250	**97–98%!**

Lesson 14: Settings Hierarchy

Level	File	Scope
User	~/.claude/settings.json	All projects on machine
Project	.claude/settings.json	Single project (shared via git)
Local	.claude/settings.local.json	Your machine, this project only

> **Precedence:** Local > Project > User (most specific wins)

4 Permission Modes:

Mode	Behavior	Use Case
default	Har tool use se pehle permission mangta hai	Learning, sensitive projects
acceptEdits	Free file read/edit; shell ke liye poochta hai	Daily development
auto	Safety classifier screen karta hai (March 2026)	Long-running tasks
bypassPermissions	Sab bina approval ke	CI/CD pipelines

Switch karne ke liye: **Shift+Tab**

Section C: Extensibility & Teams (Lessons 15–25)

Lesson 15: Hooks — Event-Driven Automation

> **"Without hooks, you RELY ON Claude remembering. With hooks, behavior is GUARANTEED."**

5 Primary Hook Events:

Event	Trigger	Use
PreToolUse	Tool execute hone se pehle	Validate, block dangerous ops
PostToolUse	Tool complete hone ke baad	Format code, run tests
UserPromptSubmit	Aap prompt submit karo	Context add, validate input
SessionStart	Claude Code launch ho	Load env variables
SessionEnd	Session close ho	Cleanup, save logs

Exit Codes:

- `0` = Success

- `2` = Action block karo with error
- Other = Non-blocking warning

Lesson 16: Plugins

Plugins = Packaged collections — skills + commands + agents + hooks + MCP servers ek saath.

```
/plugin    ← Plugin Manager khulta hai
```

Installation Scopes:

- **User scope:** Sab projects — `~/.claude/` mein
- **Project scope:** Team ke liye — `.claude/settings.json`
- **Local scope:** Sirf current repository

> **Rule:** Personal tools = User scope. Team standards = Project scope.

Lesson 17: Ralph Wiggum Loop

Autonomous iteration workflow — Claude khud task run karta hai, errors check karta hai, fix karta hai, repeat karta hai — bina user intervention ke!

> **Rule of Thumb:** "Agar 10+ iterations expect karo — Ralph Loop time bachata hai."

Architecture:

- **Stop Hook** — Claude ki normal exit intercept karta hai
- **Completion Promise** — Text jo task completion signal kare
- **Max Iterations** — Infinite loop se bacha

Best Use Cases:

- Framework upgrades (React v16→v19)
- Test-driven refactoring
- Linting/TypeScript error resolution

> **DO NOT use for: Human judgment required tasks, creative work, multi-goal tasks, tasks needing external input.**
> **Golden Rule: "Success must be objective, verifiable, and deterministic."**

Cost Guidelines:

Complexity	Iterations	Approx Cost
Simple (5-10 errors)	15-20	$10-20
Medium (10-30 errors)	30-40	$30-60
Complex (50+ errors)	50-80	$80-150

Lesson 20: Agent Teams

> **Key Distinction:** "If teammates need to talk to each other, use teams. If they just report back, use subagents."

3 Universal Coordination Patterns:

1. **Parallel Investigation** — Multiple investigators different angles simultaneously
2. **Pipeline Build** — Sequential stages, har stage ka output next ko feed karta hai
3. **Competing Hypotheses** — Multiple teams ek doosre ki theories disprove karne ki koshish karte hain

> Teams 3-5x zyada tokens use karte hain — wisely use karo!

Lesson 22: Worktrees — Parallel Isolation

Problem: Multiple sessions same files edit karein to silently overwrite!

```
claude --worktree feature-auth    ← Isolated git worktree banata hai
```

> **Gotcha:** Worktrees ALWAYS default remote branch se branch karte hain — current branch se nahi!

Lesson 23: Remote Control

Kisi bhi device (phone, tablet) se Claude Code sessions monitor aur control karo — local machine sab kaam karti hai!

```
claude remote-control    ← Start karo
/rc                      ← Existing session mein
```

> **Pro/Max/Team/Enterprise subscription zaroori. API key authentication compatible NAHI.**

Lesson 24: Scheduled Tasks

```
/loop 5m check if deployment finished    ← Har 5 min check karo
```

3 Scheduling Tiers:

Tier	Method	Restart survive?
Cloud	Cloud Scheduled Tasks	Yes
Desktop	Desktop scheduled tasks	Yes (app running)
Session	/loop skill	No

Common Cron Expressions:

Expression	Matlab
`*/5 * * * *`	Har 5 minutes
`0 * * * *`	Har ghante
`0 9 * * *`	Roz subah 9 baje
`0 9 * * 1-5`	Weekdays 9 baje

Lesson 25: Channels — Event-Driven

> **Automation Triptych:**
>
> - /loop = Time-driven (timer par)
> - Channels = Event-driven (external push)
> - Remote Control = Human-driven

Channel = MCP server with push capabilities — external systems events bhejte hain running session mein.

Section D: Claude Cowork

Cowork Kya Hai?

Autonomous agent within Claude Desktop app — local machine par kaam karta hai, approved files directly access karta hai.

> **"Cowork is an agent, not a chatbot." It takes INDEPENDENT action!**

Hardware Requirements:

- macOS: Apple Silicon (M1+) — Intel Macs sirf Chat
- Windows: x64 ya ARM64

Code vs Cowork Comparison

Aspect	Claude Code	Claude Cowork
Interface	Terminal/CLI	Desktop Cowork tab
Users	Developers	Knowledge workers
Optimal Use	Code, testing, debugging	Documents, reports, analysis
Built-in Skills	git, npm, testing	docx, xlsx, pptx, pdf
Execution	Interactive (user reviews)	Autonomous background/interactive

3 Distinguishing Cowork Features:

1. **Persistent Context** — Poore approved workspace ka awareness
2. **Document-Aware Skills** — docx, xlsx, pptx, pdf built-in
3. **Visual Feedback** — Exact changes dikhata hai execute se pehle

Computer Use

Claude aapke desktop ko control karta hai — apps open karna, cursor move, click, type!

Tool Selection Hierarchy (Priority):

1. Connectors (Gmail, Slack) — fastest
2. Bash (shell commands) — fast
3. Browser (Claude in Chrome) — medium
4. Computer Use (screen control) — slowest (fallback)

> **Claude Avoids:** Stock trading, passwords/credit card input, facial image scraping.

Section E: Strategy & Assessment

Code vs Cowork — Decision Framework

> **Core Rule: "Code for code, Cowork for documents."**

3 Decision Criteria:

1. Primary artifact type: Code files → Code; Office documents → Cowork
2. Goal: Tests run karo → Code; Batch document process → Cowork
3. User background: Developer → Code; Non-technical → Cowork

Quick Reference — All Key Terms

Term	Definition
General Agent	AI jo OODA loop se actions execute kare (Claude Code)
Custom Agent	Specific workflow ke liye optimized AI (SDK se bana)
Product Overhang	Model mein latent capability — sahi interface ka intezaar
CLAUDE.md	Persistent context file — har session auto-load
AGENTS.md	Universal AI file — 60,000+ projects use karte hain
Skill	SKILL.md file with domain expertise — portable procedures
Subagent	Isolated context window wala specialized agent
Hook	Event-driven automation — guaranteed behavior
Plugin	Packaged collection of skills+agents+hooks+MCP
Ralph Loop	Autonomous iteration plugin — bina intervention ke iterate kare

Agent Teams	Multiple Claude instances jo aapas mein baat kar sakein
Worktree	Isolated git branch environment for parallel work
Remote Control	Any device se Claude Code session control
Channels	External push notifications into running sessions
Digital FTE Formula	CLAUDE.md + Skills + MCP = Digital FTE
claude doctor	Installation/auth diagnostic command
bypassPermissions	CI/CD ke liye — sab bina approval
Completion Promise	Ralph Loop mein task completion ka signal text

Chapter 14 — General Agents | Panaversity | Exam Preparation Notes | Imran Hassan